GRT Seminar Spring 2026 – Potent Categorical Representations

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Abstract

This semester, the GRT Seminar will focus on potent categorical representations, following the 2026 ICM address of Nadler–Ben-Zvi (based on joint work with Stefanich). There are many open questions!

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1 (1/22) David Nadler – Introduction

We're going to start with the first page or so of the paper. The rough idea here is to instantiate some of the ideas in Constantin Teleman's ICM address.

1.1 Categorical representations

Our motivation for studying categorical representations comes from the following analogy.

If Γ is a finite group, then we look at the group ring $\mathbb{C}[\Gamma]$ We may think of this as a ring of distributions, with multiplication given by convolution. Studying representations of Γ corresponds to studying modules over $\mathbb{C}[\Gamma]$.

In general, if $\mathcal{C}$ is a symmetric monoidal (∞-)category, e.g. $(\mathsf{Vect}_{\mathbb{C}}, \otimes)$, we may consider an algebra object $\mathsf{A} \in \mathcal{C}$. We can study the category of modules $\mathsf{Mod}_{\mathsf{A}}(\mathcal{C})$.

We can specialize this very general framework to give a notion of categorical representations. Let G be a connected reductive group over $\mathbb{C}$.

Definition 1.1. A *categorical* G*-representation* is a (presentable) stable $\mathbb{C}$-linear ∞-category which is a module over $\mathsf{D}(\mathsf{G})$, the derived ∞-category of D-modules on G. We'll write $\mathsf{G}\text{-}\mathsf{Cat}$ for the ∞-category of categorical G-representations.

Remark 1.2. Morally, the reason we use D-modules here is because they quantize the cotangent bundle. The really important object is $\mathsf{T}^*\mathsf{G}$ (with two moment maps $\mathsf{T}^*\mathsf{G} \rightrightarrows \mathfrak{g}^*$ corresponding to the left and right actions of G). Everything here is secretly symplectic!

Example 1.3. Let $\mathsf{G} \curvearrowright \mathsf{X}$. Then we get an action $\mathsf{D}(\mathsf{G}) \curvearrowright \mathsf{D}(\mathsf{X})$. This really corresponds to the moment map $\mathsf{T}^*\mathsf{X} \to \mathfrak{g}^*$.

1.2 Categorical representations of $\mathbb{G}_m$

Let's see what happens when $\mathsf{G} = \mathbb{G}_m$.

Theorem 1.4. *There is a monoidal equivalence* $\mathsf{D}(\mathbb{G}_m) \simeq \mathsf{QC}(\mathbb{A}^1/\mathbb{Z})$.

Indeed, recall that a D-module on $\mathbb{G}_m$ comes with two pieces of data:

- An action of the Euler vector field $t\partial_t$, corresponding to the coordinate on $\mathbb{A}^1$.
- An action of t and t^{-1}, corresponding to translation-equivariance.

This correspondence sends the δ distribution at 1 to the structure sheaf.

Corollary 1.5. *There is an equivalence* $\mathbb{G}_m\text{-}\mathsf{Cat} \simeq \mathsf{QC}(\mathbb{A}^1/\mathbb{Z})\text{-}\mathsf{Mod}$.

Reasonable objects within $\mathsf{QC}(\mathbb{A}^1/\mathbb{Z})\text{-}\mathsf{Mod}$ have "singular support" in (the 2-shifted version of) $\mathsf{T}^*(\mathbb{A}^1/\mathbb{Z})$. This translates into a notion of "position" and "momentum" for such objects.

Example 1.6. If we consider a point $\lambda \in \mathbb{A}^1/\mathbb{Z}$, the module category $\mathsf{QC}(\lambda)$ corresponds to

$$\mathsf{D}_\lambda(\mathbb{G}_m) = \left\langle \mathscr{O}_{\mathbb{G}_m}, d - \frac{\lambda}{z} dz \right\rangle \subset \mathsf{D}(\mathbb{G}_m).$$

Roughly: the position of a $\mathsf{QC}(\mathbb{A}^1/\mathbb{Z})$-module corresponds to the monodromy. This relates to Langlands parameters.

Momentum should correspond to "equivariant parameters." To capture this, we must work microlocally. This is the motivation for considering potent categorical representations.

1.3 Another motivation

Sufficiently dualizable 2-categories give rise to 3d TQFTs. One can obtain some 3d TQFTs by taking S^1-compactifications of 4d TQFTs, i.e. setting $\mathsf{Z}_3(\mathrm{pt}) = \mathsf{Z}_4(S^1)$. If Z_4 is pure gauge theory, Z_3 corresponds to considering gauge theory with classical fields G/G. Remember that G/G can be thought of as the free loop space $\mathscr{L}\mathsf{BG}$. The "equivariant parameters" here live in the numerator G, while the "monodromies" / "G-representations" come from the denominator G.

1.4 Defining potent categorical representations

We can write $\mathsf{G}\text{-}\mathsf{Cat} = 2\mathsf{IndCoh}(\mathsf{G}_{\mathrm{dR}})$. Thus a possible extension is:

Definition 1.7. Let $\mathsf{G}\text{-}\mathsf{Cat}^{\mathrm{pot}} = 2\mathsf{IndCoh}(\mathsf{G}/\mathsf{G})^{\mathrm{Tate}}$.

This contains $\mathsf{G}\text{-}\mathsf{Cat}$ as the full 2-subcategory set-theoretically supported at $1 \in \mathsf{G}$. There are other subtleties (e.g. shearing, genuine equivariance) that should appear. We end up with a theorem:

Theorem 1.8. *There is an equivalence* $\mathbb{G}_m\text{-}\mathsf{Cat}^{\mathrm{pot}} \simeq 2\mathsf{IndCoh}(\mathbb{A}^1/\mathbb{Z})$.

We may also define a Lie algebra version:

Definition 1.9. Let $\mathsf{G}\text{-}\mathsf{Cat}^{\mathrm{pot,gr}} = 2\mathsf{IndCoh}(\mathfrak{g}/\mathsf{G})^{\mathrm{Tate}}$.

Our goal this semester is to understand these definitions and theorems.